North South University
Small Language Model
Course: CSE299 – Junior Design
Section: 04 | Group: 03

Team Members	Supervisor
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Weekly Update: 01
Reporting Period: 16.02.2026 – 23.02.2026

Work Completed

A. Group Member 1 – SLM Fine-Tuning & Benchmarking (MD. Adnan Abdullah Sadi)

This week focused on fine-tuning Phi-3-mini and benchmarking it against larger LLMs on mathematical reasoning tasks. Key achievements include:

- Fine-tuned Phi-3-mini (3.8B) scored 80% on GSM8K with 5-shot Chain-of-Thought (CoT), surpassing Mistral-7B (12%) and Qwen2.5-7B (40%)
- LoRA fine-tuning (SFT) delivered +20% performance improvement, taking accuracy from 56% to 72% for Microsoft Phi-3
- Hidden state embedding probes on PRM800K confirmed Phi-3 (73%) > Mistral-7B (70%) > Gemma-2B (66%), showing architecture matters more than scale
- Test on Gemma-2B's own outputs produced an honest accuracy range of 73.7%–84.7%, eliminating proxy and distribution mismatch errors
- Annotation quality analysis revealed an 11-point gap between annotators (ChatGPT: 84.7% vs Claude: 73.7%), highlighting the impact of annotation quality
- Full SLD-VM pipeline built end-to-end: LTG → RGC → VCE system ablated across 7 configurations and validated on 200 held-out problems

B. Group Member 2 – SLM Tools Performance (Makki Ammer Sakib)

This week focused on evaluating whether Phi-3-mini can effectively power real-world tools when combined with external systems. Three tools were built and tested on Google Colab using a T4 GPU:

Tool 1: Web Search Agent

- Phi-3-mini was integrated with DuckDuckGo search engine to answer real-time questions
- Phi-3 acts as the decision-maker: it decides whether to search the web or answer from model knowledge
- Finding: SLM correctly identified stable knowledge questions (e.g., photosynthesis) but gave outdated answers for current facts (e.g., Champions League titles) without search
- Solution implemented: Hybrid approach combining keyword forcing + model decision-making for better accuracy

Tool 2: Unit Converter

- Phi-3-mini extracts value and units from natural language (e.g., 'how many miles is 100 km?')

- Python handles precise mathematical conversion using a lookup table to avoid SLM calculation errors
- Phi-3 generates a natural language explanation of the result with real-world context
- Finding: SLMs are unreliable at precise arithmetic — hybrid Python + SLM approach is essential

Tool 3: Text Summarizer

- Phi-3-mini summarizes long articles in three modes: Short (2–3 sentences), Bullet Points, and Detailed
- 100% powered by Phi-3 — no external tools needed
- System prompt engineering used to control output style and length
- Finding: SLMs perform well on summarization tasks using prompt engineering alone

Overall Findings from Tool Testing:

- SLM works well for: stable knowledge, summarization, language understanding
- SLM struggles with: real-time facts, precise calculations, current events
- SLM + external tools is a powerful and cost-effective alternative to large LLMs

C. Group Member 3 – SLM Compression via MoLAE (Nahian Syed Ahanaf)

This week focused on compressing SLM feed-forward networks using SVD factorization under the MoLAE framework on GPT-2 Medium:

- MoLAE SVD-based FFN factorization applied on GPT-2 Medium with c_proj-only factorization as the core architectural contribution
- Compression targets of 5%–25% actual FFN reduction achieved with fine-tuning recovery over 5,000 steps
- Cascade compression pipeline built: MoLAE + INT8/INT4 quantization
- All 7 compression targets beat the pretrained baseline (PPL 35.83 → best 20.21)
- Novel insight discovered: c_proj-only factorization outperforms both-operator factorization
- INT4 alone catastrophically fails (PPL 159,808), strongly motivating the MoLAE cascade approach
- MoLAE + INT8 achieves ~75% size reduction while maintaining generation quality